

1 Problem & model

UWB swarm: N agents, DS-TWR ranging every $T = 100$ ms; snapshots G_t are 3-D random geometric graphs (range R_c)

Routing: OLSR (RFC 3626) computes routes on the *advertised* graph $\hat{G}_t$

Threat model: 10–20% insiders (A1, A2, A9, A10a/b, A12–A14) with black/grey-hole dropping; external PHY attacker (A4 reduce, A5 enlarge)

2 Critique

Faithful reproduction of ranging-integrated MPR (GRR, Eq. 3 of Zhou et al.)

compared with the ILP optimum, flooding and RFC 3626 (simulator and ns-3)

Findings O1–O3: near-flooding relay sets; –17–18 pp vs. RFC 3626; damage lies in route computation

3 GeoTrust+ (every TC interval, at each agent i)

TESLA-authenticated ranging units (8 B CMAC)

Kinematic gate $g = 2v_{\max}T + 4\sigma$

Reciprocity disagreement graph: own-link blame, vertex cover (degree ≥ 2), one-sided claims

One-sided loss clamp $\hat{P} \geq P(d) - \tau_P$

Overhearing watchdog binomial test, $z < -3.5$

Conviction set X_i (local, never gossiped)

RFC 3626 MPRs on $\Gamma_1 \setminus X_i$ + quarantine-aware next hops

4 Security analysis

P1: PHY distance enlargement cannot frame honest agents

P2: an undetected liar gains at most $\tau_P + L\tau_R$ in loss

Collusion, framing by omission, watchdog limits; cost $O(|\Gamma_1|^2)$ per interval

5 Evaluation

Calibration attack-free runs, seeds disjoint from evaluation; $\tau =$ quantile $1 - \alpha/4$, $\alpha = 0.01$

Swarm simulator (Python/GPU) sparse ($R_c=2$ m) and dense ($R_c=4$ m) swarms of 50 agents; 9 attacks + PHY; N , speed, obstacles; ablation and threshold sensitivity; 30 seeds per cell

ns-3.48 (packet level) 802.11g + OLSR, 30 drones; HELLO-lie hook and emulated ranging; 7 attacks, 20 seeds (720 runs)

MCU (Cortex-M4F, C port) bit-exact with the simulator; QEMU instruction counts; DWT benchmark for boards

6 Network science

Statistics & outputs mean $\pm 95\%$ t-CI, paired Wilcoxon tests; open artifact (Zenodo)

Structure: swarm RGG vs. Erdős-Rényi graphs — clustering, auditability, betweenness concentration, link churn

Betweenness capture delivery loss vs. fraction of flows routed through insiders

Targeted placement random vs. top-degree vs. top-betweenness insiders